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(COURSE DESCRIPTION)

DATE: 2026/09/08
INDEX: 27001

Zanjan Girls Technical and Vocational College - Zanjan

Associate Degree
Computer - Computer software

Remarks:
According to the Academic Instruction of University, in the academic system, each theoretical credit is presented in 16 or 17 hours, each practical credit in 32 or 34 hours,each of the workshop credit in 48 or 51 (except Professional Certificates) and each internship or training credit in 64 to 120 hours.

There are some different type of the courses passing by a student at this university, as listed below:

1. Basic
2. Core / Major
3. Specialized
4. Elective / Optional
5. General
6. Internship / Workshop
7. Project / Thesis

Each one of these types contain some courses to be passed, according to the educational regulations passed by the Ministry of Science, Research and Technology or Ministry of Health & Medical Education of Iran.

The grading system in this university is from 0 to 20. The minimum passing grade for a course leading to an Associate's Degree or a Bachelor's Degree is 10, for a course leading to a Master's Degree and Medical Sciences is 12 and for a course leading to a PhD Degree is 14.

Note: This document has been prepared at **Zahra Koosha's** request in the **Course Description** Platform. Scan the QR-code below to certify this document.

www.coursedescription.ir

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University Info :

Address : Zanjan, Zanjan Province, Iran
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Course	Credits	Description
Advanced programming 1	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized)** The purpose of this course is to study the principles and methods necessary to produce a computer program with good quality specifications. In this regard, after covering the top-down design method to solve the problem, students will be introduced to the concepts and techniques of object-oriented programming as a tool for managing complexity in medium and large-scale applications. Program performance, testing, and debugging are focused throughout the course. An object-oriented programming language (such as Java) is used to teach these principles and methods, and students are introduced to the concepts and techniques of object-oriented programming and module-based object design, advanced concepts such as graphical user interface design. , Multidisciplinary programming (parallel) and distributed programs on the network are also covered in this course. **Syllabus :** In depth Java programming, Introduction of UI design QT. Complementary issues of Java programming, Memory management, in depth understanding of Java codes, Coding relation with operating system, file management, IO streams, clear implementing basic data structures like link lists, Generic programming, Implementation of inheritance and its related issues in Java. Multithreading essentials, exception handling, object oriented programming principles, Debugging and testing of programs, Function calling conventions, Dynamic memory coding.

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Advanced programming 2	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized)** This course focuses on advanced programming concepts and methodologies, emphasizing practical skills for designing and implementing complex software systems. Students will deepen their understanding of algorithms, data structures, and modern programming paradigms, including object-oriented and functional programming. The course integrates theoretical knowledge with hands-on practice using professional tools and frameworks. **Syllabus :** Advanced Data Structures. Graph algorithms and network flow. Balanced trees and segment trees. Hashing techniques and collision resolution. Modern Programming Paradigms. Advanced object-oriented programming concepts. Functional programming techniques and practices. Event-driven and asynchronous programming. Software Development Frameworks. Introduction to widely-used software frameworks. Application development with APIs and SDKs. Debugging and optimization in integrated development environments. Algorithm Optimization. Performance analysis and complexity reduction. Parallel and distributed algorithm implementation. Memory and resource management techniques. Advanced Programming Tools. Version control systems and collaborative coding. Static and dynamic code analysis tools. Automated testing frameworks and CI/CD pipelines. Programming in Real-world Applications. Implementation in domains like data science, AI, or web development. Handling scalability and security challenges in code. Software deployment and maintenance practices.
Internet Basics	1	**(Duration: 32 hours Practical)** **(Type of Course: Specialized)** This course has two sections: Internet and Windows. Internet Section The Internet section is designed to provide students with an introduction to the Internet including, email basics and advanced features, web browser basics, web navigation, web research and newsgroups. Windows Section The Windows section is designed to provide students with an introduction to the current version of Microsoft Windows Professional or Windows XP. Students will learn to manipulate the Windows environment, use Windows Accessories and use My Computer and Windows Explorer to manage files and folders.

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Operating system 2	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Specialized)** Study of advanced operating system topics with a focus on infrastructure technologies for enabling cloud computing, including virtual machines, containers, resource management, system software reliability and verification, security, mobile computing, and distributed systems. Course will involve readings and discussion of classic and new papers on recent developments in operating system research. Course will develop understanding of operating systems and distributed systems as well as project management, research methodology, and technical presentation and writing skills.
Operating system workshop 2	1	**(Duration: 32 hours Practical)** **(Type of Course: Specialized)** This Lab provides enriching practical skills and imparting industry-relevant advanced operating system concepts to students and faculty members in the field of Information Technology. This Lab provides a platform to learn the knowledge, practical ability, innovation and system design capability, Networking, Information Access, Distributed Operating systems, Collaborative Computing, Computer Graphics, and operating system design language.
Database	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Specialized)** It includes representing information with the relational database model, manipulating data with an interactive query language (SQL) and database programming, database development including internet applications, and database security, integrity and privacy issues.
Database Lab	1	**(Duration: 32 hours Practical)** **(Type of Course: Specialized)** Understanding the fundamentals of database management systems **Syllabus :** Introduction to database concepts. Data modeling and database design. Relational databases and SQL. Normalization techniques. Indexing and performance optimization. Query optimization. Transaction management and concurrency control. Database security and backup strategies.
Local Computer Networks	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Specialized)** Understanding the fundamental concepts and principles of computer networks. **Syllabus :** Introduction to networking, network topologies, network protocols, OSI model, TCP/IP protocol suite, Ethernet technology, routing and switching, network security, wireless networks, network troubleshooting.

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Local Computer Networks Workshop	1	**(Duration: 48 hours Practical)** **(Type of Course: Specialized)** Understanding the principles of computer networks and gaining practical skills in setting up and managing local area networks (LANs). **Syllabus :** Introduction to network devices (routers, switches, etc.), TCP/IP protocol suite, network topologies (star, bus, ring, etc.), subnetting, IP addressing, VLANs, network security concepts, network troubleshooting techniques, introduction to network monitoring tools.
Data Structure	3	**(Duration: 48 hours Theoretical)** **(Type of Course: Specialized)** Familiarity with Information Structures, Effects of Structures in Produced Applications, Selecting Optimized Inside-Memory Structures, Organizing Memory based on requirements. **Syllabus :** Arrays, vectors, matrices, private matrices, arrays display, stacks, queues and rows, bond lists: graphic, cyclical, double bond, multi-bond, method of display and application of bond lists, trees and their menstruation, method of representation and application of trees: decision making trees, search trees, tree of the game and etc., graphs and their representation, dynamic memory allocation and the relevant issues, searching and sorting and combining algorithms
Information Storage and Retrieval	3	**(Duration: 48 hours Theoretical)** **(Type of Course: Specialized)** Information retrieval is the process of responding to a user's request for textual information related to a particular topic. Information retrieval is one of the most important and primary issues in the field of natural language processing. One of the most important applications of information retrieval is web search, in which most people have experienced working with information retrieval systems. The purpose of this course is to acquaint students with the basics, models, tools and applications of information retrieval. **Syllabus :** Definition of storage and retrieval, study of types of extra-machine memories (disc, tape and ...) regarding calculation of real rate of transfer and principal parameters, basic concepts of file, blocking techniques, calculation of factor of block and wasted memory in them, buffering and the relevant techniques and efficiency of each one, study of different structures of file and different operation on each of them, evaluation criteria of different structures (the rate of consumed memory, operation time), analytical description of each of the unordered, ordered, indexed, ordered-indexed, multi-index, direct and multi-loop structures, external sorting and merging, study and implementation of one of the advanced access methods.

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Web-based programming	2	**(Duration: 32 hours Practical & 16 hours Theoretical)** **(Type of Course: Specialized)** This course provides an introduction to web development and client-side scripting. After providing a review of HTML5 and CSS, the course provides exposer to the concepts of web programming using client side scripting. The course covers basic construction of web page, cascading style sheet, and java script.
Machine and Assembly Languages	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized)** Reminding use of different units of computer, definition of register, introduction of its types and its application, concept of instruction format in machine language, types of instructions, different stages of instruction execution, methods of addressing (implied , immediate, direct, indirect, indexing, base, relocation, related to content of program counter, paging).Concept of relocation, instruction by registers , instruction by memory (manipulation of half word full word, bytes and bits), jump instructions and control of loop , logical instructions, dexinal manipulation , floating point manipulation subprograms, recursive subprograms, macros, trace and debugging of program, damping introduce assembly and its types.
Special Topics	2	**(Duration: 32 hours Practical & 16 hours Theoretical)** **(Type of Course: Specialized)** Teaching and discussing a new topic in the field of study with the opinion of the faculty and the supervisor.
Multimedia environments	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Specialized)** This course focuses on how new computer technologies for learners can be created and used in an educational setting. The following topics will be explored: understanding the learner as a technology user, defining learning outcomes to be supported by technology, differing approaches to the technology design process and methods of technology integration in the classroom. In the first half of the semester, students will read current publications on technology design methods and educational theories. During class sessions, the readings will be discussed in relation to small group design projects. During the first part of the semester, student teams will develop new multimedia environments for learners using the web, video, and hypermedia.

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Fundamentals of Software Engineering	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Specialized)** The purpose of this course is to address the engineering points that must be observed in all stages of software production. In this course, students are introduced to the activities and tools needed to produce a software product. **Syllabus :** Importance and objectives of software engineering; planning; stages and design stages controlling; principles of needs analysis; methods of needs analysis; principles of software design; study of several known methods of design such as data structure, data flow, functional, aimed and immediate analysis, outstanding points of programming languages in viewpoint of software engineering, software accuracy and confidentiality, software maintenance, performance of a practical project.
Principles of supervision	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Specialized)** This four-day, CSU-designed, Principles of Supervision training program covers basic supervision skills to include performance management, developing direct reports, guiding and organizing the work, managing relationships, and managing yourself. **Syllabus :** Introduction to Supervision. The Role of a Supervisor. Communication Skills in Supervision. Motivation and Employee Engagement. Decision Making and Problem Solving. Conflict Resolution and Mediation. Performance Evaluation and Feedback. Time Management and Delegation. Ethical Considerations in Supervision. Diversity and Inclusion in the Workplace. Team Building and Collaboration. Change Management and Adaptability. Emerging Trends in Supervision. Case Studies and Practical Applications. Final Project and Presentation.

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Entrepreneurship	3	**(Duration: 48 hours Theoretical)** **(Type of Course: Specialized)** The purpose of this course is to teach the basic concepts of entrepreneurship and creating new and knowledge-based businesses. In this course, students get to know the principles of business plan formulation, company launch, marketing and sales. **Syllabus :** Concepts and definitions of entrepreneurship, evolution of entrepreneurship in the world, importance and necessity of entrepreneurship. Definition of innovation and creativity, innovation process, levels and types of innovation, principles of invention and innovation, idea and opportunity. Organization and structure of a business, types of organizational structures, steps of business management, duties of a business manager, familiarity with types of companies, steps of setting up knowledge-based companies and leadership in business. Familiarity with company regulations, business rules, brand registration, trademark and licensing procedures. Management and timing of work and interests in business, financial management, definition of financial activities, capital estimation, capital sources, preparation of company's financial documents and documents, asset management, profit and loss account and expenses. Familiarity with marketing and sales, marketing tools, definition of advertising and the purpose of advertising, use of media and advertising tools, digital marketing in the future world, international business (trade and export) and entrepreneurial culture. Business model preparation, model difference and Business plan, how to set up and present a business plan, familiarity with growth and entrepreneurship centers and capital acquisition methods, familiarity with the concept of employability
Internship	2	**(Duration: 240 hours Practical)** **(Type of Course: Specialized)** Students must practically get involved in industry and market by working totally 240 hours in an appropriate and defined company in either governmental or private sectors as an intern.
Project	2	**(Duration: 96 hours Practical)** **(Type of Course: Specialized)** Evaluation of the student's ability to use the specialized courses of the field of study **Syllabus :** Conducting preliminary studies, collecting materials, designing experiments, making devices, preparing data and analyzing them, drawing conclusions and presenting reports.

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Graphic Software Lab	1	**(Duration: 32 hours Practical)** **(Type of Course: Core)** This is an introductory course on principles of computer graphics. We will consider both 2D and 3D graphics. Broadly speaking, we will look at raster scan graphics including line and circle drawing, polygon filling, anti-aliasing algorithms, clipping, hidden-line and hidden surface algorithms including ray tracing and, of course, rendering - the art of making photo realistic pictures with local and global illumination models. Implementation of basic and advanced algorithms will be done in OpenGL and C++. Basic knowledge of C/C++ programming is mandatory.
Electronics Fundamentals	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Core)** The course also provides a light introduction to the actual circuit design, construction, test, and measurement, giving the hobbyist a jumping-off point for further pursuit of the subject. In addition, the course highlights some key concepts of semiconductors and semiconductor devices (like transistors). Finally, the lessons conclude with a look at some applications of the principles discussed throughout the course. The lessons are designed to serve students with a variety of backgrounds, and they require only a minimal level of mathematical aptitude.
Electronic Basics Workshop	1	**(Duration: 32 hours Practical)** **(Type of Course: Core)** This course is an introductory experimental laboratory that explores the design, construction, and debugging of analog electronic circuits.
Computer Hardware 2	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Core)** Introduces the operation of a modern personal computer from a hardware point of view. Students learn about the basic components of a computer and develop troubleshooting skills for advanced courses. Some of the areas covered include system boards, storage drives, memory and power supplies. In addition, some software topics, such as operating systems, are presented.

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Written and spoken presentation methods	1	**(Duration: 16 hours Theoretical)** **(Type of Course: Core)** **Syllabus :** Different types of scientific and technical subjects (letters, reports, pamphlets, manual and etc.), common points in all scientific and technical writings: specifying the objective of writing and its eventual readers, organizing the subjects, abstract of essay together with report, the role of a good introduction, dividing the subjects into parts and chapters, discussion and conclusion, preparing source and reference index, attachments, preparing the pictures and diagrams and tables. Important points in translation of scientific and technical subjects, writing style, marking and its importance, preparing final format of writing by typing machine or computer, foot-article, notes and other lateral subjects, an introduction to research methods, presenting subjects orally, effective use of audio-visual devices.
Technical language	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Core)** This technical course provides trainees with the skills and technical vocabulary to discuss effectively a broad range of technical issues with colleagues and clients. Trainees will use authentic texts and realistic tasks to help them prepare for their demanding environment.
Pre-university Mathematics	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Compensatory)** Introduces fundamental math concepts. The topics include whole numbers, operations on whole numbers, fractions, decimals, ratios, rates, and proportions. The focus is on learning the computational procedures and then applying the skills to problem solving in applications.
Statistics and Probability	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Core)** Identify the types of data (qualitative, quantitative, discrete, and continuous). Identify the types of sampling (random, stratified, systematic, cluster). Identify the misuses of statistics. Student will use appropriate statistical methods to collect, organize, display, and analyze relevant data. **Syllabus :** Topics include probability axioms, conditional probability, the law of total probability, Bayes' theorem, independence, discrete and continuous random variables, multiple random variables, sum of random variables, the sample mean, and introduction to statistical inference.

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General Mathematics 1	3	**(Duration: 48 hours Theoretical)** **(Type of Course: Basic)** Basic concepts of Calculus and Geometry will be taught to the students in this course which provides necessary background for technical courses. **Syllabus :** Cartesian coordinates; polar coordinates; complex numbers; addition, product, root & geometrical representation of complex numbers; polar representation of complex numbers; function; functions algebra; limit and relevant theorems; infinite limit and limit in infinite; left-hand and right-hand limit; connectivity; derivative; derivation formula; inverse function and its derivative; trigonometric functions derivative and their inverse functions; Rolle's theorem; mean theorem; Taylor expansion; geometrical and physical applications of derivative; curves and acceleration in polar coordinates; application of derivative in approximation of equations roots; definition of integral of continuous functions and piecewise continuous; basic theorems of differential & integral arithmetic; primitive function; approximate methods of integral estimate; application of integral in computation of area, length of curve, moment, center of gravity and labor ....(in Cartesian and polar coordinates); logarithm and exponential function and their derivative; hyperbolic functions; integration methods such as change of variable, component and decomposition of fractions; transform of special variables of sequence and numerical series and relevant theorems; power series and Taylor theorem with remainder.

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General Mathematics 2	3	**(Duration: 48 hours Theoretical)** **(Type of Course: Basic)** After completing the course, the student will find the appropriate mental background to understand and solve various engineering problems and acquire the ability to calculate the vectors, determinants, matrices, partial derivatives, complete differentials, and spherical and cylindrical coordinates. **Syllabus :** Parametric equations; space coordinates; vector and space; numerical product; matrix 3x3 of three-indeterminate linear equations system; operation on lines; matrix reverse; solving equations system; linear independence; base in R2; R3 linear transform and its matrix; determinate 3x3 and characteristic value and vector; vector product; second order line and plane equations; two vector functions and its derivative; speed and acceleration; bending; normal vector to a curve; multivariable function; directional and partial derivative; tangent plane and normal line to a curve; multivariable function; directional and partial derivative; tangent plane and projecting line of gradient; chain of rule for partial derivative; exact differential; second kind and third kind integrals and their application in geometrical and physical problems; transform of integration arrangement (without accurate affirmation); cylindrical and spherical coordinates; vector field; curvilinear integral; surface integral; divergence; curl; Laplacian; potential of green space and divergence and stochastic.
Applied Mathematics	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Core)** This course focuses on mathematical principles and methods essential for solving practical problems in applied contexts. Topics include quantitative analysis, modeling, and advanced techniques for optimization and financial computations. **Syllabus :** Two-variable functions, limit and continuity, twofold integral, differential equations, Solving first-order equation ,Solving second-order equations, Solving second-order linear homogeneous equations with constant coefficients, Solve the Applications of Electrical Circuits, Laplace transform, Calculating Functions Using the Laplace's Laws, Calculation of sinusoidal functions using Laplace transform, Calculation of integral and derivative functions using Laplace transform, calculation of First and Secondary Differential Equations, Sequence and series.
Physics of electricity and magnetism	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Basic)** This course is an introduction to electricity and magnetism and their mathematical description, connecting electric and magnetic phenomena via the special theory of relativity. Topics include electrostatics, magnetic fields, electromagnetic induction, DC and AC circuits, and the electromagnetic properties of matter.

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Foreign (English) Language	2	**(Duration: 32 hours Theoretical)** **(Type of Course: General)** The content of this course is presented based on the acquisition of skills in reading and understanding the concepts of foreign texts in order to prepare students for the placement of semi-specialized texts. **Syllabus :** A review of the structures and vocabulary of reading different texts. Emphasis on learning skills including: A- How to use English dictionary and encyclopedia. B - Vocabulary. C - Writing outlines. D - Summarizing. E- Fast reading different texts.
Persian language	3	**(Duration: 48 hours Theoretical)** **(Type of Course: General)** Familiarity with Persian language and reading literary texts, poetry and prose in Iranian Persian literature.
Physical Education	1	**(Duration: 32 hours Practical)** **(Type of Course: General)** Physical Education classes are designed to practice and develop skills in activities that will help you maintain fitness throughout your life. During the year, you will set and monitor personal fitness goals that will aid in the development of a personal fitness plan.
Family and Population Knowledge	2	**(Duration: 32 hours Theoretical)** **(Type of Course: General)** Familiarity of students with the basics of family knowledge and how to form, consolidate, and excel it and demographic characteristics and their role in improving the quality of life
Applied Ethics	2	**(Duration: 32 hours Theoretical)** **(Type of Course: General)** Familiarity of students with the moral standards of Islam about the most important aspects of daily life and creating a context for a greater tendency to accept the moral standards of Islam on individual and social issues.
Islamic Thought (Origin & Resurrection) 1	2	**(Duration: 32 hours Theoretical)** **(Type of Course: General)** The objectives of this course include studying the concept of origin and the concept of the afterlife in Islamic philosophy.

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